

Orator — Website Maintenance Report

Automated review and fix pass · 19 June 2026

Branch: **claude/relaxed-sagan-9wamfb** · Repo: mickhatzikostas220-design/speech-analyzer2

Summary

I reviewed the Orator Next.js application end to end, building and linting the project to surface real problems. The headline finding was a **build-breaking bug**: the production build failed outright. I fixed that and a cluster of related correctness, security, and quality issues. After the changes, both **npm run build** and **npm run lint** complete cleanly. All work is committed and pushed to the feature branch.

Changes at a glance

Severity	Area	Change
Critical	Build	Lazy-init OpenAI client so the production build no longer crashes when secrets are absent at build time.
High	Rendering	Marked auth-gated (app) and (auth) layouts force-dynamic so per-user pages are never statically cached.
High	Security	Closed an auth gap: /compare was reachable without a session; now protected in middleware.
Medium	Security	Documented the undocumented ADMIN_ACTION_SECRET and switched admin-token verification to a timing-safe compare.
Medium	Tooling	Added ESLint config + typescript-eslint so lint runs non-interactively; fixed every surfaced error.
Low	Branding	Unified stray 'ACA' references (navbar logo, chat prompt) to 'Orator'.

1. Build failure — module-level OpenAI client (Critical)

Files: lib/openai.ts, app/api/compare-report/route.ts

The OpenAI client was constructed at module load time (`const openai = new OpenAI({ apiKey: process.env.OPENAI_API_KEY })`). During Next.js page-data collection the module is evaluated, and with the key absent the constructor throws, aborting the entire build with “The OPENAI_API_KEY environment variable is missing or empty”.

Fix: the client is now created lazily on first use inside the request path, and the compare-report route is marked *force-dynamic*. Secrets are a runtime concern, not a build-time one.

2. Static prerendering of authenticated pages (High)

Files: app/(app)/layout.tsx, app/(auth)/layout.tsx

The shared Navbar instantiates the Supabase browser client during render, and the server pages read per-user session data. Next.js was attempting to statically prerender these routes at build time, which both failed without env vars and risks serving cached, user-specific pages in production.

Fix: both layouts now export `const dynamic = 'force-dynamic'`, forcing on-demand server rendering for every dashboard, history, compare, editor, and auth page.

3. Unprotected /compare route (High, security)

File: middleware.ts

The auth middleware redirected unauthenticated visitors away from /dashboard, /analysis, /history, /admin and /editor — but /compare was omitted from the protected list despite being linked in the navbar and loading user analyses. An unauthenticated user could reach it directly.

Fix: added `path.startsWith('/compare')` to the protected-path check so it redirects to /login like the other app routes.

4. Admin action secret & token verification (Medium, security)

Files: .env.local.example, lib/adminToken.ts

ADMIN_ACTION_SECRET signs the one-time approve/deny links emailed to admins, but it was never documented and falls back to a hardcoded 'fallback-secret-change-me' if unset — meaning anyone could forge valid action links against a default deployment. It is now documented in the env example with an explicit warning.

Token verification also compared HMAC signatures with a plain `!==`, which leaks timing information. It now uses a length check plus `crypto.timingSafeEqual`.

5. Lint setup and code-quality fixes (Medium)

Files: .eslintrc.json (new), package.json, and several pages/components

The project had no ESLint configuration, so `npm run lint` dropped into an interactive prompt and could not run in CI. I added a config extending next/core-web-vitals and @typescript-eslint/recommended (installing the plugin/parser), with no-unused-vars honoring the underscore convention.

I then fixed every error it surfaced: unescaped quotes/apostrophes in JSX (editor, history, signup, request-access, success pages), an unused import and several unused variables in the compare page, and dead code (getVal, METRIC_DESCRIPTIONS, scoreColor). Two remaining items are `<img>` warnings on dynamic video frames, intentionally left as-is.

6. Branding consistency (Low)

Files: components/Navbar.tsx, app/api/analyses/[id]/chat/route.ts

Two stray references called the product “ACA” — the navbar logo and the chat assistant's system prompt — while the rest of the app, metadata, and README use “Orator”. Both now say “Orator”.

Verification

- ✓ **npm run build** — compiles successfully, all 19 pages generate.
- ✓ **npm run lint** — no errors (only two pre-existing warnings on dynamic frames).
- ✓ All changes committed and pushed to **claude/relaxed-sagan-9wamfb**; a draft PR was opened.

Notes for follow-up

- The @anthropic-ai/sdk dependency is declared but unused in source — a candidate for removal if not planned for use.
- npm audit reports vulnerabilities in transitive deps; a separate dependency-bump pass is advisable.
- README “Known limitations” (no rate limiting, placeholder waveform, approximate Tribe ROI indices) remain open by design.